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Department of Physics, Durham University
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A table-top probe of wafer bonding: AFM and nanoscale gas drainage

Drainage — the process by which a liquid or gas is removed from the gap between approaching solid surfaces — is central to direct wafer bonding, adhesion, and self-assembly. Two polished silicon wafers brought into contact seal together through van der Waals interactions and capillary condensation along a travelling bond front, which sweeps radially at centimetres per second and squeezes out trapped gas from a gap that closes to sub-nanometre proximity. The wafer-scale bond event couples long-range elastic plate bending and an autonomous front-speed selection to a local drainage law in the closing nanometre gap; that local law is the closure every continuum lubrication model inherits (Jain, 2026; Navarro et al., 2013; Rieutord et al., 2005). The continuum description breaks down as the gap closes below ~ 100 nm, where the Knudsen number approaches unity. A generalised lubrication equation with Knudsen-slip corrections is the natural candidate for that sub-continuum regime, but it has not been benchmarked against a controlled single-asperity experiment. That benchmark is the project. A heavily blunted atomic force microscope (AFM) tip, pressed against a polished substrate under controlled humidity, delivers a single-asperity realisation of the local drainage law; axisymmetric direct numerical simulations of the same geometry close the loop.

TL;DR

You will run AFM force-distance experiments (with Dr Kislon Voïtchovsky, Durham) using a heavily blunted tip pressed against a flat substrate under a humidity sweep from dry nitrogen ($< 5\%$ RH) upward, and you will build direct numerical simulations (with Dr Vatsal Sanjay, CoMPhy Lab) of the squeezed film between tip and substrate. The two sides meet at a quantitative test of the nanoscale drainage constitutive law in the $\mathrm{Kn} \sim 1$ slip regime — the same sub-continuum piece that controls void formation and unbonded area in industrial hybrid bonding at imec and TSMC (Nagano et al., 2022; Sato et al., 2025). This is the minimal model: the local drainage law, isolated from wafer-scale bending elasticity, that a lubrication solver based on Rieutord–Navarro theory needs as closure input.

Description

Direct wafer bonding is the process by which two polished semiconductor wafers are brought into contact and seal together through van der Waals interactions and capillary condensation (Kim et al., 2018), with the ambient gas expelled radially as a bond wave propagates from an initial contact point (Gösele et al., 1995; Masteika et al., 2014; Plößl and Kräuter, 1999). The established thick-substrate front-speed scaling, derived by Rieutord and collaborators for flexible silicon wafers separated by a thin fluid layer, is $v \propto w^{5/4}$ with w the wafer thickness (Rieutord et al., 2005); the most complete published model couples plate-bending theory to Reynolds lubrication and an adhesion-front boundary condition (Navarro et al., 2013). A recent reformulation for flexible substrates recovers that thick-substrate limit and exposes a distinct thin-substrate / near-edge regime in which $U \propto \gamma^{3/2} E^{-1/2} T^{-3/2}$, driven by capillary stress alone without viscous dissipation in the leading order balance (Jain, 2026); that framework stays within continuum lubrication-elasticity and explicitly excludes adhesion mechanisms and conjoining/disjoining pressures. The industrial figures of merit are not overlay alone (overlay is a downstream alignment metric) but *void density* and *unbonded area* (Nagano et al., 2022; Sato et al., 2025). Each of these is controlled by what happens inside the closing gap. As the gap approaches the molecular mean free path ($\lambda \approx 68$ nm for air at STP), Knudsen

slip, rarefaction, and sub-continuum gas dynamics all enter, and a continuum Reynolds closure drifts. Real closing gaps live at Knudsen numbers $\mathrm{Kn} = \lambda/h \sim \mathcal{O}(1)$: a regime that is neither continuum nor free-molecular, and where the appropriate closure is the Fukui–Kaneko generalised lubrication equation tabulated from DSMC-validated Poiseuille coefficients (Fukui and Kaneko, 1988, 1990).

The bond-wave physics is hard to probe on a production scanner — the gap is buried between two 300 mm wafers and closes in milliseconds. An AFM single-asperity geometry is the minimal model by construction: stripping out wafer-scale bending and autonomous front propagation isolates the same nanoscale squeezed-film law that a continuum bond-front model takes as its closure. A standard AFM cantilever has a tip radius of $\mathcal{O}(10)$ nm; deliberately blunting it (Butt et al., 2005) to $\mathcal{O}(1\text{–}10)\,\mu$m produces a hemispherical punch of known radius R, which can be driven against a polished substrate at controlled approach velocity U under controlled humidity (Butt and Kappl, 2018). The force-distance curve carries three separable signatures: (i) pre-contact capillary jump-in at humidity-set separations (Israelachvili, 2011), (ii) squeezed-film viscous resistance that follows Stefan's law $F_{\mathrm{visc}} = 3\pi\eta R^2 U/(2h)$ (sphere-on-flat, thin-gap) at low Kn but drifts predictably as slip enters at $h \lesssim \lambda$, and (iii) a pull-off force set by JKR or DMT adhesion (Maugis, 1992), modulated by any condensed bridge formed during contact. The AFM sweep maps the local constitutive law $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ — where $\mathrm{Ca} = \eta U/\gamma$ — not the front itself. The mapping to industrial bonding is indirect but concrete: once the local drainage law is quantified, it can be embedded as the closure in a lubrication solver based on Rieutord–Navarro theory, and it distinguishes the thick-substrate $U \propto \gamma^{5/4} E^{-1/4} T^{-3/4}$ regime from the flexible-substrate / near-edge $U \propto \gamma^{3/2} E^{-1/2} T^{-3/2}$ limit (Jain, 2026). The AFM geometry itself lacks the wafer-scale bending stiffness, so the classical thick-substrate scaling is not expected to apply directly to the cantilever-on-flat measurement; the experiment probes the local constitutive law that feeds both regimes.

Placeholder for two-panel schematic: (a) wafer bond-wave geometry (bending plates + lubricating gas layer + advancing adhesion front), (b) blunted-AFM single-asperity geometry isolating the nanoscale squeezed-film law that feeds (a).

Figure 1: The AFM is a local constitutive probe, not a full bond-wave analogue. Panel (a) shows the wafer-scale problem: two bending plates, a lubricating gas layer, an autonomously propagating adhesion front. Panel (b) shows the single-asperity AFM geometry that isolates the thin-gap drainage law $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ entering (a) as a closure.

This project closes that local loop. On the experimental side, you will run force-distance curves with a blunted AFM tip on silicon and oxide-terminated substrates, sweeping both approach speed (four decades, 10^{-9} to $10^{-5}\,\mathrm{m\,s^{-1}}$) and relative humidity (from $< 5\%$ dry-nitrogen purge upward in steps, so gas drainage and capillary condensation are separable in the data, not conflated (Butt et al., 2005)). On the theoretical side, you will build axisymmetric direct numerical simulations in Basilisk C (Popinet, 2015; Sanjay, 2022b) of the closing gap, with the Fukui–Kaneko slip-corrected lubrication closure for the gas phase to cover the continuum-to-rarefied transition (Fukui and Kaneko, 1988, 1990), and a disjoining-pressure potential $\Pi(h)$ with $\int \Pi\, dh = W_{\mathrm{adh}}$ as the molecular-adhesion closure. The experiment and the simulation meet at one non-trivial prediction: the speed and humidity dependence of the squeezed-film force at $\mathrm{Kn} \sim 1$. The tested closure then feeds back into lubrication models based on Rieutord–Navarro theory.

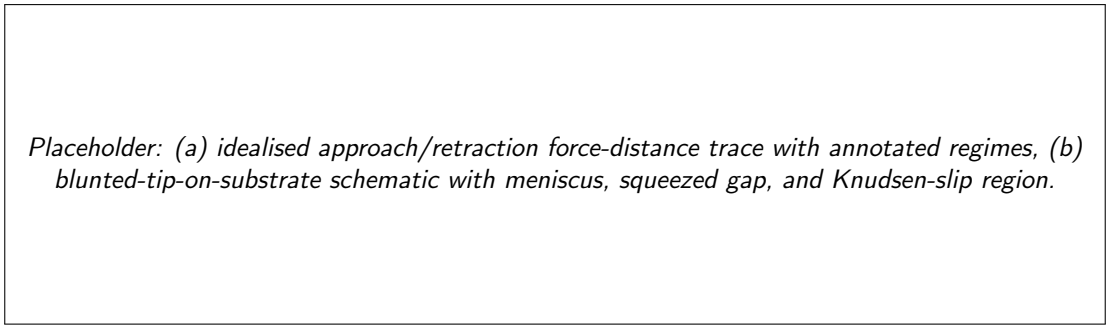


Figure 2: AFM force-distance signature of the three-force balance. Capillary jump-in (high RH, low U), viscous drainage (high U, continuum), and pull-off (adhesion-dominated retraction) are separable regimes of the same underlying model.

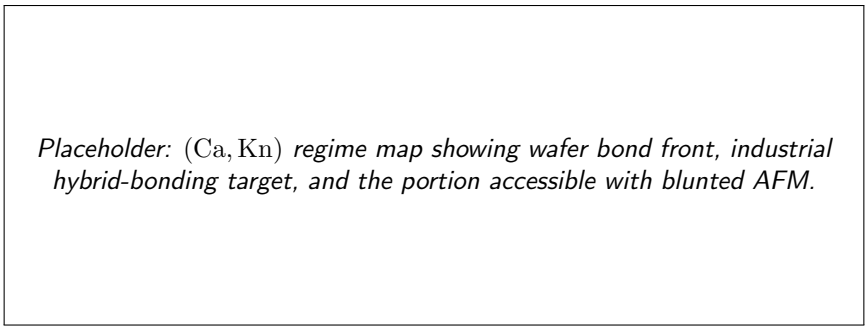


Figure 3: Dimensionless map. The blunted-AFM sweep covers the industrially relevant Ca-Kn corner where the continuum lubrication description breaks down and sub-continuum gas physics sets the bond-front velocity.

What you will do and what you will learn?

1. Learn force-spectroscopy AFM in Dr Voïtchovsky's lab: cantilever calibration, tip blunting protocols, humidity-controlled force-distance measurements, and data reduction from raw deflection to force.
2. Build an axisymmetric Basilisk C simulation of a blunted-tip approach to a flat substrate, with a slip-corrected Reynolds layer for the gas phase and a meniscus model for condensed water. Follow the CoMPhy-Lab two-phase and documentation conventions (Sanjay, 2022a,b).
3. Derive the local squeezed-film law in the three canonical limits — continuum Stefan ($\mathrm{Kn} \to 0$), Knudsen slip ($\mathrm{Kn} \sim 1$), and capillary-bridge dominated (high RH, low U). Test the Fukui–Kaneko generalised lubrication closure (Fukui and Kaneko, 1988, 1990) against the force-distance data, identify where it drifts, and bracket its quantitative validity range in Kn. Place the measured $F(h; \mathrm{Kn}, \mathrm{Ca}, \mathrm{RH})$ alongside both wafer-scale front-speed scalings — the thick-substrate limit $U \propto \gamma^{5/4} E^{-1/4} T^{-3/4}$ (Rieutord et al., 2005) and the flexible-substrate / near-edge limit $U \propto \gamma^{3/2} E^{-1/2} T^{-3/2}$ (Jain, 2026) — treating the AFM data as a direct measurement of the local drainage closure that lubrication solvers based on Rieutord–Navarro theory require as input.
4. Position the results in the wafer-bonding landscape through the primary literature: the Rieutord 5/4 derivation (Rieutord et al., 2005), the Navarro coupled plate-bending / lubrication model (Navarro et al., 2013), the flexible-substrate reformulation of Jain that exposes the

additional $\gamma^{3/2}E^{-1/2}T^{-3/2}$ near-edge regime and deliberately leaves adhesion and conjoining/disjoining pressures outside the continuum lubrication-elasticity closure (Jain, 2026), the Gösele–Tong bond-speed analysis (Gösele et al., 1995), Masteika's review (Masteika et al., 2014), and the Nagano and Sato process-yield studies that actually quantify the industrial figures of merit (Nagano et al., 2022; Sato et al., 2025). Write a short section of the final report on how the measured local drainage closure feeds a lubrication solver and which sub-100-nm physics it fixes for the wafer-scale problem.

5. Develop the CoMPhy-Lab full-stack open-source skillset: Basilisk, adaptive mesh refinement, Python post-processing, GitHub documentation, and writing for a combined soft-matter and semiconductor-physics audience.

Industrial and scientific context

Hybrid bonding is the enabling technology for 3D-stacked logic, high-bandwidth memory, and backside power delivery in the sub-2 nm node. TSMC's SoIC and Intel's Foveros both use it in volume. The dominant yield-limiting failure modes reported in the public process-engineering literature are not overlay error but *void formation* and *unbonded area* (Nagano et al., 2022; Sato et al., 2025); overlay is the downstream alignment metric that these upstream failures corrupt. Each of these failures is seeded inside the closing gap, where sub-continuum gas drainage at Knudsen numbers around unity sets whether the front sweeps cleanly or pockets gas. Existing continuum-Reynolds closures in lubrication solvers based on Rieutord–Navarro theory (Navarro et al., 2013; Rieutord et al., 2005), and the flexible-substrate lubrication-elasticity extension of Jain (2026), do not cover that regime; the Fukui–Kaneko generalised lubrication equation does, but it has not been directly benchmarked against a controlled single-asperity experiment in the relevant Kn range. The blunted-AFM setup is a minimal model with controlled boundary conditions: it isolates the local closure and feeds it back to the wafer-scale solver. For a Durham MPhys or Master's project, the outputs are the measured local drainage law at $\mathrm{Kn} \sim 1$, the matching axisymmetric simulations, and a short discussion of how that closure feeds wafer-scale lubrication solvers based on Rieutord–Navarro theory and which sub-100-nm physics it fixes.

If you have any questions, feel free to contact us vatsal.sanjay@comphy-lab.org/ vatsal.sanjay@durham.ac.uk or drop by Ph255 (Rochester building) at the Department of Physics at Durham University.

Collaborators	E-mail	Based at
Dr. Vatsal Sanjay	vatsal.sanjay@comphy-lab.org vatsal.sanjay@durham.ac.uk	Ph255 (Rochester building)
Dr. Kislon Voïtchovsky	kislon.voitchovsky@durham.ac.uk	Dept. of Physics, Durham
Dr. Utkarsh Jain	utkarsh.jain@imec.be	imec, Leuven

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